

Canadian Math Olympiad 2005 — P2/5

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Let (a, b, c) be a Pythagorean triple, i.e. a triplet of positive integers with $a^2 + b^2 = c^2$.

(a) Prove that $\left(\frac{c}{a} + \frac{c}{b}\right)^2 > 8$.

(b) Prove that there are no integer n and Pythagorean triple (a, b, c) satisfying $\left(\frac{c}{a} + \frac{c}{b}\right)^2 = n$.

Solution

For part (a) we have

$$\frac{c^2}{a^2} + \frac{c^2}{b^2} + 2\frac{c^2}{ab} = \left(1 + \frac{b^2}{a^2}\right) + \left(1 + \frac{a^2}{b^2}\right) + 2\frac{a^2 + b^2}{ab} \geq 1 + 1 + 2 + 4 = 8$$

due to AM-GM. But since $a = b = c$ is impossible the inequality is strict.

For part (b) we notice that $\frac{c}{a} + \frac{c}{b}$ must be an integer because if not then $n \notin \mathbb{Z}$. Consider the primitive Pythagorean triple (a', b', c') , that is we divide a, b and c by $\gcd(a, b, c)$. Now similarly $\frac{c'}{a'} + \frac{c'}{b'} = \frac{c'b' + c'a'}{a'b'}$ must be an integer. If $a'b' \neq 1$ then $a'b' \mid c'$ but this is impossible since $\gcd(a', b', c') = 1$. Otherwise if $a'b' = 1$ then $a' = b' = 1$ and $c' = \sqrt{2}$ which isn't an integer, so contradiction. Thus indeed there is no such integer n . ■